

Pandas EDA Practice Questions

Dataset: ecommerce_sales.csv | 20 Questions | Exploratory Data Analysis with Pandas

About the Dataset

The file **ecommerce_sales.csv** contains 1,012 rows of simulated online retail order data. It intentionally includes real-world messiness — missing values, duplicate rows, inconsistent category labels, and a handful of outliers — so you can practice a full exploratory data analysis (EDA) workflow: loading, cleaning, transforming, and analyzing data with pandas.

Column	Description
order_id	Unique identifier for each order
order_date	Date the order was placed (YYYY-MM-DD)
category	Product category (e.g. Electronics, Clothing)
product_name	Name of the product ordered
region	Sales region (North, South, East, West, Central)
city	City the order was shipped to (some missing values)
customer_age	Age of the customer (some missing values)
customer_segment	New, Returning, or VIP customer
payment_method	Payment method used (values are inconsistently formatted)
unit_price	Price per unit in USD (contains a few outliers)
quantity	Number of units purchased
discount_pct	Discount percentage applied to the order
total_sales	Final sale amount after discount
rating	Customer rating 1-5 (some missing values)
delivery_days	Days taken to deliver the order (contains outliers, some missing)

Instructions: Load the dataset with `pandas.read_csv()` and answer each question below. Show your code and output. Questions are grouped by theme and increase in difficulty within each section.

Part 1: Data Loading & Inspection

Q1. Load the dataset and display the number of rows and columns, along with the data type of each column.

Q2. Display the first 10 rows and the last 5 rows of the dataset.

Q3. Check for missing values in each column and report the count and percentage of missing values per column.

Q4. Check whether the dataset contains any fully duplicated rows. If duplicates exist, remove them and report how many rows were removed.

Part 2: Data Cleaning

Q5. The 'payment_method' column has inconsistent formatting (e.g. 'Credit Card' vs 'credit card', 'Cash on Delivery' vs 'COD'). Standardize this column so that equivalent values are grouped together, then show the cleaned value counts.

Q6. Handle the missing values in 'customer_age' and 'delivery_days' using an appropriate strategy (e.g. median imputation). Justify your choice in a short comment.

Q7. Handle the missing values in the 'city' column. Decide whether to drop these rows or fill them with a placeholder such as 'Unknown', and explain your reasoning.

Q8. Identify outliers in 'unit_price' using the IQR (interquartile range) method. List the order_ids of the rows flagged as outliers.

Part 3: Descriptive Statistics

Q9. Generate summary statistics (mean, median, std, min, max, quartiles) for all numeric columns using describe().

Q10. What is the average, minimum, and maximum 'total_sales' across the entire dataset?

Q11. What is the average customer rating, and what percentage of orders have a rating of 5?

Q12. Calculate the average delivery time in days, and identify which orders took longer than 14 days to deliver.

Part 4: Grouping & Aggregation

Q13. Group the data by 'category' and calculate the total and average 'total_sales' for each category. Which category generates the most revenue?

Q14. Group the data by 'region' and find the number of orders and total revenue per region. Which region has the highest number of orders?

Q15. Using a pivot table, show the average 'total_sales' broken down by both 'category' and 'customer_segment'.

Q16. Group by 'payment_method' (after cleaning it in Part 2) and calculate the average order value and average rating for each payment method.

Part 5: Time Series & Trend Analysis

Q17. Convert 'order_date' to a proper datetime type, then create a new column for the order month (e.g. '2024-03'). Show the number of orders per month.

Q18. Identify which month had the highest total sales revenue across the full dataset.

Q19. Calculate total sales by day of the week (Monday through Sunday). Is there a noticeable weekday vs weekend pattern?

Part 6: Relationships & Deeper Analysis

Q20. Investigate whether there is a relationship between 'discount_pct' and 'quantity' purchased. Use grouping and/or a correlation coefficient to support your conclusion.

Q21. Segment customers by 'customer_segment' (New, Returning, VIP) and compare their average order value, average rating, and average delivery time. Summarize your findings in 2-3 sentences.

Tip: Useful pandas methods for this workbook include `info()`, `describe()`, `isna()`, `duplicated()`, `groupby()`, `pivot_table()`, `value_counts()`, `astype()`, `dt` accessor, and `corr()`.